Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/15/2005
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Project Title:	Integrated Biomimetic Sensors Using Artificial Hair Cells		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Other		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	61801	**Congressional District:**	15
Comments:	HQ BITR		
Project Type:	Ground	**Solicitation / Funding Source:**	OTHER
Start Date:	01/01/1999	**End Date:**	08/01/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	3
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			

Highly advanced sensory perception capability is a critical requirement in future space exploration and development. To be able to empower human operators by performing dexterous and autonomous tasks in unstructured environments, robotic systems must possess large numbers and variety of sensors under relatively high density of integration. It is also critical that sensors be developed in an extremely agile and cost-effective manner. The state of the art of modern sensors is not sufficient to satisfy this challenging requirement due to long development cycle, high costs, and zero adaptability.

The innovative objective of this research is to develop an entirely new methodology for sensor development based on a biological inspiration, the hair cells. Honed by millions of years of evolutionary selection, hair cells in biological systems have become highly sophisticated. A most important aspect of biological inspiration lies in the fact that single biological hair cell serves as a common building blocks for an astonishingly wide variety of sensory functions used in a large number of living systems; a variety of functions are realized by varying the parameters of hair cells and by array

configuration. This aspect offers a new paradigm compared with current engineering practices.

Task Description:

Inspired by this fundamental biological sensor principle, we will develop a new type of sensor called the artificial hair cell, by leveraging recent advancement in microfabrication technologies and integrated sensors. We plan to conduct research to demonstrate a new sensor-development methodology, that of building functionally varied sensors using simple and common units, at much reduced time and cost. The work will potentially realize a scale of economy that has never existed in the sensor field, and benefit future NASA applications with high performance, low cost, adaptable sensors with wide-ranging functions. We focus on the development of a new type of sensor called the artificial hair cell, by leveraging recent advancement in microfabrication technologies and integrated sensors. We plan to (1) develop design and micromachining processes for realizing such sensors; (2) characterize the performance of such sensors. Through this work, we plan to develop the fundamental design, fabrication technology and materials understanding for realizing artificial haircell sensors, which are conjectured to be able to reduce the cost of sensor systems for future NASA applications. A new sensor concept is being introduced that can potentially satisfy application needs of NASA, military, and civilian industry. The research work will lead to new discoveries for low cost micromachined sensors in general.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Accomplishments
Major accomplishments are summarized in the following according to several major areas:

Development Area 1: Development of Artificial Haircell Sensors Dr. Liu's group successfully developed a novel microfabrication process combining bulk silicon micromachining and three-dimensional assembly to realize an artificial haircell array based on single-crystal silicon. The vertical cilium was developed using plastic deformation magnetic assembly method, which is very efficient and capable of generating arrayed devices over large wafer surface areas. Force applied horizontally to the cilium – including fluid flow, touch, and vibration – created torque acting on the vertical cilium. The torque translated to the horizontal cantilever and generated stress at the location of in-plane strain sensors. The amount of applied force could be inferred from the change of resistance.

Later Dr. Liu's group had further advanced the state-of-the-art by realizing an artificial haircell sensor based on photosensitive polyimide. Polymer material is of lower cost and greater durability than silicon material. The polymer based artificial haircell, schematically diagrammed in Fig3a, is composed of a vertical beam (artificial cilium) rigidly attached to the substrate. Attached at the base of the beam, between the cilium and the substrate, is a strain gauge. The strain gauge is comprised of a thin film Nichrome (NiCr) resistor on a thicker polyimide backing that runs the length of the cilium. When an external force was applied to the vertical beam, either through direct contact with another object (functioning as a tactile sensor) or by the drag force from fluid flow (flow sensing), the beam will deflect and cause the strain gauge to stretch or compress. The strain gauge region was treated as being rigidly attached to the substrate, while the cilium is free. The magnitude of the induced strain was greatest at the base, where the strain gauge was located.

All polymer micro sensors are desired in certain cases, such as flow sensing. Previous work with artificial haircell (AHC) MEMS flow sensors [1] and elastomer patterning [2] has been combined to realize a wholly polymer flow-sensing structure. Newly developed fabrication processes combined unorthodox techniques (e.g., lost wax sacrificial processing) to realize tall (3 mm) and high aspect ratio vertical polyurethane integrated with precision-patterned elastomer force sensitive resistors (FSRs). Although conductive rubber FSRs have been used in robotics for years [3] and commercial FSR products exist, these sensing elements have not been previously photolithographically patterned.

Dr. Liu's group conducted a detailed study investigating the properties and performance of 2-part polyurethane rubber elastomers as materials for MEMS. So far, polydimethylsiloxane (PDMS) is the most commonly used elastomer in MEMS, especially microfluidics. But PDMS has limited mechanical properties. Polyurethanes are stiffer than PDMS and offer better thermal degradation and tear resistance. His group also demonstrated new designs and processes for polyurethane polymers with a robust, all-polymer artificial haircell sensor.

The group established a new material for the MEMS field. Polyurethanes represent a group of elastomeric polymers with wide-ranging properties. Like PDMS, polyurethanes offer low cost and mold process capability. Characterization of a full set of material and processing-related properties (e.g., surface treatment, tensile modulus, adhesive strength, chemical resistance, and thermal degradation) has been performed based on two polyurethane rubbers (PMC121 and PMC790 [4]). Thermal degradation testing showed that polyurethanes withstand long-term exposure to temperatures commonly encountered in MEMS processing, with weight loss greater than 1% beginning at 125ºC. Tensile testing showed that the modulus of PMC121 polyurethane (453 kPa) is close to that of PDMS (750 kPa for Sylgard-184), while PMC790 is an order of magnitude stiffer (19.1 MPa). In addition, the elongation at break of both polyurethanes is well over 200% strain and both exhibit notch-blunting behavior while PDMS will readily crack. The principle of operation of the all-polymer sensor is discussed below. The carbon-filled urethane FSRs arranged in a pair of half bridges can determine x and y-axis deflection of the 3mm tall cilia with on and off axis performance. On-axis is the response from x-axis FSRs to x-deflection; off-axis is the response from x-axis FSRs to y-deflection and vice versa. Tip deflection sensitivity is increased 20-fold over earlier work (245ppm/µm vs. 12.3ppm/µm). The current prototype designs do not exhibit perfect off-axis insensitivity due to non-linear behavior of the FSRs and non-uniformity in FSR baseline resistance. The polyurethane hairs can be repeatedly bent by 90o.

Area 2: Development of Artificial Lateral Line Flow Sensing System Biological haircell sensors not only offer highly efficient sensing, but also high performance signal processing and sensor networking as well. Signal processing of artificial haircell sensor involves analog to digital conversion and amplification. In biological haircell, these electronics functionalities are achieved with a single neuron. Even with today's advanced microelectronics circuit technology, the signal processing circuitry still occupies much greater footprint than a neuron. Although it is difficult to probe single neuron functionality, we examine the neuron response to mechanical stimulations to identify features of biological sensor design and networking.

The second generation artificial haircell sensor has been combined with hotwire anemometers to form a sensor pixel, which is directly integrated on top of semiconductor integrated circuit substrates with existing signal processing units. In

fact, the hot wire anemometer offers better sensitivity than the haircell sensors at this stage and was used primarily for flow sensing and visualization. The hot wire sensors are described below. In the past, efforts have been made to apply various micromachining processes to realize HWA for measuring airflow with smaller dimensions, better uniformity, and faster time response. One strategy is to use the bulk-micromachining technique to produce freestanding cantilever structures [5, 6]. Another group undercut a polysilicon hot-wire with wet etching so the hot-wire is suspended over a cavity for better thermal insulation [7]. However, the consumption of the silicon substrate in bulk-micromachining increased the difficulty for integration with electronics circuits. Dr. Liu's group monolithically integrated the out-of-plane flow sensor on a silicon substrate with pre-existing circuitry. The sensors were fabricated on top of commercial JFET quad operational amplifier (LF347) wafer made by National Semiconductor. Each hot-wire was connected in a feedback bridge circuit to a separate amplifier die to maintain overheat (1-R/R0) of 0.25. The sensors are repeated on each die across the array. The fabrication was as follows: First, a 5-µm-thick polyimide was spun on to passivate and planarize the circuit and subsequently patterned to expose the metal contacts. This was followed by a series of surface micromachining steps to deposit the aluminum sacrificial layer, polyimide structural layer, nickel hot-wire, gold wiring, and Permalloy flap. The aluminum sacrificial layer was then removed and an external magnetic field was applied to plastically deform the gold hinge and raise hot-wire out of plane. At the same time, selective electroplating of nickel reinforced the outer two hinges. A picture of a fabricated array shows the dashed line is the size of a single die. Two types of sensors were fabricated in the same process, the hot-wire anemometer and an artificial hair cell. Each set of sensors occupies an individual die separated 2.3 mm apart.

The response of several hot-wire to airflow showed that the voltage swing is about 100mV over a 20m/s flow speed. With the calibration curve we can translate the output voltage swing into airflow. Using a standard data acquisition software, we mapped the voltage output of the fabricated hot-wire array for flow imaging. A flow imaging data showed at time t=0, the peak flow is on the right side of the array, and from 0>t>18seconds, the fan is swept across the array. The region of high flow speed (represented by color red) is move from the right side to the left side in the image.

Last year, Dr. Liu's group focused their efforts on two issues: 1) Demonstrate functionalities of artificial haircell sensors for flow sensing applications. The group focused on developing flow sensors integrated with on-chip electronics signal processors in order to image flow fields (liquid or air). The flow sensor for liquid sensing is modeled after the fish lateral line system. (The lateral line system is described in more details below).

2) Further develop polymer micromachining processes and incorporate new polymer materials such as polyurethane. The neuromast themselves are relatively complicated organs made of several components. Each neuromast typically contain tens to hundreds of hair cells, which are individual nerve cells that form the fundamental building block of the octavalateralis system, and are found in all vertebrates. They are inherently displacement detectors that have a preferred orientation to give them directionality. They haircells that populate a neuromast are always arranged in two groups that intermingle with each other but have directly opposite orientation. Each neuromast is innervated by at least 2 afferent nerve fibers, one for each direction. Multiple nerve fibers can connect to the same neuromast. For canal neuromast, a nerve fiber never connects to more than one neuromast. Similarly, for superficial neuromast, a nerve fiber will not connect to neuromast that are in different rows. The extremity of the neuromast is covered by a gelatinous sack called the cupula. The cupula serves to protect the hair cells and to couple motion of the surrounding fluid to haircell deflection. It has a density that is similar to the surrounding fluid, and is driven primarily by viscous force[14, 15]. The cupula displacement is proportional to the velocity of the surrounding fluid.

The actual sensory unit found in the octavolateralis system is the haircell. Protruding away from the haircell are bundles of hair called the cilia. They are arranged in order of increasing height, with the shortest cilia at one end, and the tallest one (kinocilium) at the other end. The top of the stereocilia are connected by tip links. When the haircell are deformed in the direction towards the kinocilium, the tip link experiences tensile stress, which leads to the opening of the ionic transduction channel into the haircell and increases in the firing rate of the nerve fiber. Deflection in the opposite direction away from the kinocilium leads to the closing of the ionic transduction channel[16]. This understand of haircell transduction is described as the gating-spring model[17]. Haircell is very sensitive and has been estimated to sense motions down to 0.3nm range, below where Brownian motion becomes significant. The signals from the two opposing direction of haircells are conveyed to the nervous system in separate channels. In canal neuromast, the haircells typically orient along the direction of the canal. Lateral line hair cell differ in dimension across species. Using incident light polarization microscopy, the haircell found in ruff, Acerina cernua L, has been visualized in vivo[18]. The haircells are pear shaped and 15µm high. The base dimension is about 7~10µm.

Micromachined hot-wire lateral line system The design and fabrication of the new type HWA is based on surface micromachining. Micromachining describes the manufacturing process of creating features with critical dimension the µm range. Many, but not all, of the process used are derived from the technologies developed for integrated circuits (IC) manufacturing[19]. Like IC manufacturing, multiple devices can be fabricated on each die, and multiple dies on each wafer. Therefore, hundreds or thousands of HWA can be fabricated simultaneously with good yield and uniformity. The fabrication process involves surface micromachining, a additive and subtractive process common in microelectromechanical systems (MEMS) to create free-standing cantilevers[20]; as well as a three-dimensional assembly process that can raise the cantilevers out of plane. Surface micromachining is used to define the HWA, including a nickel-iron alloy support prong, and a nickel-polymer composite hot-wire sensing element. 3D assembly is used to raise the sensor out of plane, which thermally insulate the hot-wire and allow the sensor to interact with the fluid above the boundary layer. Detailed fabrication and discussion has been presented in detail[1] and will not be discussed in detail here, but it has been demonstrated that large array of the sensor can be fabricated in parallel for flow sensing, and even monolithically integrating with basic signal conditioning and amplifying circuits[21].

A schematic of a single micromachined HWA shows the hot-wire is different with the conventional ones, it is made of a thin film nickel/polyimide composite. The 100 nm-thick-nickel is deposited by thermal evaporation with an as-deposited sheet resistance of 1O/ , and has a temperature coefficient of resistance (a) of 4100ppm/°C. The nickel filament is sandwiched by 2 layers of polyimide that serves to provide passivation, structural support, and electrical isolation. This has an advantage of achieving a large electrical resistance due to thin-film metal and reduced conduction end loss due to the low thermal conductivity of polyimide. HWA fabricated in this method can have wire length from as small as 50 to as long as 400µm, and as tall as 1mm. For the single channel testing presented here, a 200-µm-long hot-wire elevated 1mm high is used. For the array testing, because we want a higher sensitivity, a 400-µm-long hot-wire elevated 600 µm is used.

In the experiment, an artificial lateral line system was fabricated using micromachined, out-of-plane, HWA to serve as individual neuromast. Each HWA array (die) was made of 16 individual sensors equally spaced 1mm apart. The sensors

were fabricated on a 2” silicon wafer that can hold up to 8 dies. After surface micromachining process, the dies were separated and the HWA was assembled using our custom 3D assembly process. Each array was then packaged and wire-bonded onto a PC board and waterproofed by using a 2µm thick vapor-coated parylene film that waterproof all the exposed wiring and to prevent electrolysis of water at the hot-wire.

Artificial Lateral Line Canal packaging

As noted in earlier discussion, the superficial neuromast response primarily to viscous drag from the difference in velocity between the animal and the surrounding fluid. When the velocity difference is large the output signal superficial neuromast can be swamped with this DC noise and become relatively insensitive to the small ac stimulus. One solution is to follow the example from biology and imbed the sensor in an artificial lateral line canal, which serves as a high-pass filter to attenuate the DC signal. To do that, we add an additional molding step similar to the method we used to create a water channel for volumetric flow rate measurement[21]. A silicone elastomer, polydimethylsiloxane (PDMS), is mixed in a 5:1 ratio and then poured onto a Teflon mold patterned by milling. The PDMS is then cured inside the mold and then separated. Some holes are punched into the PDMS to create the similar pore structure and the PDMS channel is then optically aligned with the HWA array and permanently bonded. The artificial canal is a useful package because it also can protect the out-of-plane HWA from being damaged.

Area 3: Characterization of Artificial Lateral Line Sensing System For the individual hot-wire measurement, the HWA was interfaced with a modular signal conditioning system from TSI that includes a anemometer module (model 1053B), a decade resistance module (model 1056) to control the overheat ratio, and a power supply module (model 1051-1). The output passes through a Grass Telefactor P55 AC preamplifier, with the band pass filter set between 10 and 300Hz, and a gain of 60dB. For array measurement, a custom circuit (circuit for a single channel) was used for signal conditioning for all 16 channels on the HWA that includes a constant temperature (CT) feedback circuit whose output was sent to an instrumentation amplifier. A trimming potentiometer is used to remove the DC offset before each measurement.

Single channel testing

All testing of the HWA was performed in water. The setup is similar to the hydrophone and neural recording experiment[22]. A B&K 4010 minishaker was attached to a 3 axis linear stage, with one axis of motion controlled by a warm gear motor. A 3mm diameter sphere was attached to the minishaker through a 12-gauge needle that serves as the dipole source. An accelerometer was attached to the base of the needle to measure the acceleration of the dipole. The packaged HWA was mounted on a test fixture that is rigidly attached to the base of the water tank through a suction cup. Since this is a single sensor channel calibration, the TSI anemometer circuit is used along with the Grass preamplifier. The signal amplitude was measured using the HP 3581 waveform analyzer.

The input-output response of the HWA to dipole amplitude shows the center of the dipole source is positioned 4.5mm away from the HWA, voltage supplied into the shaker is attenuated by the value shown, so the larger the attenuation the lower the applied voltage. The reference value of the attenuation is 0.7 Vrms at 40dB. As the applied voltage to the shaker drops, the acceleration of the dipole source decreases linearly. Because the HWA is not directional, we expect the doubling of frequency so the amplitude from the two harmonics at 1× and 2× the frequency. The slope of the 1× harmonic is –1, like that of the accelerometer. However, for the 2× harmonic, the output initially falls off at –1, but falls off at –2 slope at higher attenuation between 50-60dB. Dubbleday attributed this phenomenon to free convection at low signal amplitude[23], but that is not the case here because axis of dipole field is parallel to gravity. Another explanation could be the imperfect symmetry of the polyimide-nickel-polyimide composite forming the hot-wire. Next, the response of the HWA to changing dipole frequency is examined. The HWA is first calibrated for its thermal frequency response in water. The frequency response of the hot-wire itself is measured by placing the HWA in constant-circuit mode at various overheat. The HWA is placed in a bridge without the feedback, and R1 is set so that for the bridge to be in balance, the HWA has to reach a particular overheat. The hot-wire –3dB cutoff frequency near 400Hz. When the HWA operates in the CT mode, the dynamic range extends to 4kHz. This shows that the HWA can respond to the low frequency dipole field in the tens to hundreds of hertz range that aquatic animals commonly encounter.

The response of a single HWA to dipole field is measured by moving the dipole stimulus across the sensor using the setup described in hydrophone experiment[22]. For a single HWA, the TSI anemometer module is used and subsequently amplified by 60dB. The HWA is set to an overheat of 0.1, corresponding to a average temperature increase from ambient of 25°C. The dipole stimulus maintains the same frequency and amplitude while traversing across the sensor and the output of the HWA is sent into an A/D module and the rms voltage of the HWA is measured.

The RMS amplitude of the digitized HWA response is plotted along with the magnitude of calculated pressure gradient. The two curves were shifted so the peak corresponds to right below the dipole source at position zero. The amplitude for both plots was normalized to the maximum value so the peak is at 100% response. The noise level of the HWA sensor was also measured as the dipole traverse across the sensor, but with the minishaker turned off. It is represented by the dotted line, and the amplitude corresponds to about 4% of the peak amplitude.

The measured data has good correlation with the model when the sensor is near the dipole source, around where the main peak is located, and far away from the source. However, around 5-20mm from the center where the side lobe should be, which corresponds to a 180° phase shift of the sinusoidal motion, the data from the sensor does not correlate well with the model. One explanation could be the output of the sensor near the side lobe (between 7-20mm away from source) almost overlaps the noise floor, which may completely overlap the side lobes. However, the sensor amplitude at the peak is 20 times the noise floor, and the model prediction shows the side lobe should be at 20% of the amplitude. The model is normalized to a particle displacement of 3µm. The noise floor corresponds to about 4% of that, corresponding to a measurable particle displacement threshold of approximately 0.1µm. This value is similar to the lowest displacement reported by Coombs[24]. Dipole sweep was done in both directions, and the sensor output is the same in each direction. One drawback of using the HWA, as discussed before, is its lack of directionality. Therefore, only the RMS amplitude comparison was performed.

Array measurement In the previous experiment, one sensor was used to resolve the dipole field by collecting the data as the dipole source traverse across it. In the real system, it is desirable to simultaneously collect data to locate the dipole source. The artificial lateral line chip with 16 HWA channel is used.

The 16-channel HWA array was interfaced with a custom 16-channel constant temperature circuit along with amplifier. The bridge ratio in the constant temperature circuit was set to 1, and the amplifier gain to 500. The HWA array used had a 400µm long by 10µm wide hot-wire, with an average resistance of 42.8O ±3% deviation. Overheat of the 16 channel

Task Progress:

HWA was adjusted using 16 individual potentiometers (R1) to an overheat of 0.06, corresponding to an average delta T of 15°C. The conservative overheat value used is to prevent the accidental burn-out of any sensor in the array. Typically, HWA used in liquid is limited to an overheat of less than 0.1 due to concerns with bubble-formation that can cause errors in the measurement[25].

The output of the HWA circuit was sent to a 12-bit 16 channel data acquisition card running LabView. Although a 16 channel sensor array is fabricated and tested, data from only 15 channels are actually collected, the final channel records the waveform from the accelerometer. A typical waveform captured is plotted, with the shaker turned on at t=0. The waveform shows a slight DC shift occurs for the first second, this transient could be due to some thermal mixing. To avoid this effect on the measurement, a pause was introduced between the start of the dipole and the start of the data acquisition. The procedure is as follows. First, the motor moves the dipole into position, and then the minishaker starts and creates the dipole field. At the same time a TTL trigger tells the data acquisition software to wait for about 2 seconds, after which data acquisition starts and captures a segment of the overall waveform until approximately 100ms before the minishaker stops. This process is repeated until the dipole source has traversed across the entire range. Each segment of the waveform captured corresponds to a particular sensor channel and dipole location. The composite of the waveform can be compiled together to piece together a spatial response curve.

Particle motion that is dominated by near-field dipole flow should fall off very quickly, as a function of 1/r3, as predicted. The attenuation of the sensor signal to distance from the source is plotted, along with a curve fit line showing that the signal does attenuate at that rate. The same data is also plotted versus the calculated particle displacement based on the time derivative. The dipole velocity U0 used to calculate the particle displacement is derived from the accelerometer output, and is measured to be around 1m/s with approximately 3mm dipole displacement. The noise floor of the HWA array is measured in absence of any dipole stimulus, and is around 10mVrms, as represented by the line. This corresponds to a minimal detectable particle displacement of around 10µm, which is considerably worse than the 100nm threshold measured for the single HWA channel interfacing with a commercial anemometer circuit and amplifier. However, it is cost prohibitive to build such a complicated circuit for all 16 sensor channels.

The RMS amplitude of the 16 channel array is plotted with the dipole source 5, 6, and 7mm above the array at two different frequencies (50 and 75Hz). The direction of the oscillation is parallel to the axis of the array. As in the single channel response, the HWA data corresponds very well near the peak, but not near the side lobes. Although it is apparent that the side lobe are beginning to form near the right most channel at position 5. The magnitude of the response drops off very quickly, and is also captured by the sensor array. To verify that all 16 channel is working, and also to show the ability of the artificial lateral line sensor array to track the dipole source, the dipole source is traverse along the axis of the sensor array. The output of the HWA array at various dipole source location is plotted, showing that the sensor array can track the motion of the dipole source. The main peak is imaged by the array as it traverses across.

HWA directionality

The polyimide-nickel-polyimide hot-film is not completely symmetric. The top polyimide layer is about 20~25% thicker than the bottom polyimide layer due to limitation in curing. This means that after the sensor is actuated, there may be some asymmetry in the temperature profile. The thicker polyimide will face one direction along the axis of the array and reduce the sensitivity, and the thinner layer will face the opposition. This was observed by looking at the ac waveform of the sensor.

The dipole response from the 16 channel HWA is plotted when the dipole source is located right above sensor at the 8mm position (position 8). The output of the two sensors to the left of the main peak (6 and 7) is plotted in the waveform on the bottom left, and the output of the two sensors to the right of the main peak (9 and 10) is plotted in the waveform on the bottom right. The frequency doubling characteristics is apparent in the waveform plot because the HWA should be a rectified waveform of particle displacement. For all HWA there exist a waveform peak at both the maxima and the minima of the accelerometer. Because of the thermal asymmetry, the two peaks in the HWA output are not the same. For both set of plots, the larger peak is labeled with an asterisk. For the two sensors to the left of the dipole, the larger peak coincides with the minima of acceleration (maxima of the particle displacement). For the two sensors to the right of the dipole, the larger peak coincides with the maxima of the acceleration. The thicker polyimide layer faces the dipole for one set of sensors, and the thinner layer for the other set of sensors. These partial asymmetries suggest some signal processing algorithm can be used to tell the directionality of the particle displacement. The output is in phase with the accelerometer and suggesting that the anemometer responds to particle displacement and not velocity, which would be 90° out of phase with the accelerometer signal. The imposed DC flow due to natural convection is estimated to be on the order of 1mm/s[23], which gives an O of around 5 according to , in the range where the HWA should be displacement sensitive.

Area 4: New Materials for Bioinspired Sensors Traditional MEMS processes are based on silicon materials, which are brittle and cannot sustain large deformation or impact. Biological haircells are made of organic materials that are extremely flexible and yet provide for high sensitivity. In this work, the researchers realized that to achieve high performance of sensors and to reduce the cost of making sensors, new materials are important and should be introduced. A conscious effort was made to incorporate new materials, especially organic materials, and to develop processing methods based on these materials. Polydimethylsiloxane (PDMS) is the most commonly used elastomer in MEMS, especially microfluidics. Although available in an array of formulations, PDMS's have limited mechanical properties, balanced by good chemical and thermal degradation resistance. In comparison, polyurethanes can be either mechanically softer or stiffer than PDMS, offer better adhesion and tear resistance, and are available in biocompatible formulations that exhibit excellent chemical resistance. Polyurethane elastomers represent a group of polymers with wide-ranging properties. Polyurethanes have found use in micro-technology as encapsulating structures, chemical sensors [26], electromagnetic absorbers [27], wells for studying blood cells [28], and artificial gecko hairs [29]. In the case of the gecko hairs, polyurethanes were used to mold sub-micron features. Characterization of surface treatment, tensile modulus, thermal degradation, chemical resistance, and adhesive strength has been performed based on two polyurethane rubbers (PMC121 and PMC790 [4]). Contact Angle Similar to PDMS, the polyurethanes tested exhibit increased contact angle (more hydrophilic) with oxygen plasma treatment. Samples were cast and cured according to manufacturer instructions, and then placed in 300W planar oxygen plasma at 500mT for 30 seconds. From the table it is clear that the polyurethanes (+25° on average) do not exhibit the same dramatic change in contact angle as PDMS (+82°). Thus these polyurethanes do not have a clearly defined and thus useful change from hydrophobic to hydrophilic surface character.

Characterization of Mechanical Properties Tensile testing showed that the modulus of PMC121 polyurethane (453 kPa)

is similar to that of Sylgard-184 PDMS cured with 10:1 resin to curing agent (750 kPa [30]), while PMC790 polyurethane is an order of magnitude stiffer (19.1 MPa). In addition, the elongation at break of both polyurethanes is well over 200% strain, with PMC121 approaching 500%. Both polyurethanes also exhibit extensive notch-blunting behavior while PDMS readily cracked when deformed. Thus the polyurethanes have low notch-sensitivity and are tougher. Room temperature vulcanizing (RTV) silicones not traditionally used for microfluidic biocompatible systems have been formulated that exhibit lower notch sensitivity as well. Thermal Degradation Thermal degradation testing showed that polyurethanes withstand long-term exposure to temperatures commonly encountered in MEMS processing, with weight loss greater than 1% beginning at 125ºC. Sylgard-184 did not exhibit 1% weight loss until 235ºC. Color can be used as an indicator for the thermal degradation of the polyurethanes, beginning with light amber and progressing to a dark orange at temperatures above 200ºC.

Chemical Resistance

Placing samples of PMC790 polyurethane and Sylgard-184 PDMS in a number of different reagents, we found that the polyurethanes exhibit marginal long-term resistance to organic solvents, with good resistance to acids and bases. Chemicals used included DI water, acetone, isopropyl alcohol (IPA), toluene, buffered hydrofluoric acid (10:1) oxide etch (BHF), 2:1 phosphoric acid (HPO4), 2:1 hydrochloric acid (HCl), and 40wt% potassium hydroxide (KOH) solution. The PDMS samples exhibited a similar trend, with better resistance to solvents. This data represents the average of three test samples.

Adhesion

The post-cure and cast-in-place adhesion of Sylgard-184, PMC121, and PMC790 to various substrates was measured, with and without the addition of a commercial adhesion promoter, using 90º peel tests. The stiffer PMC790 exhibited superior adhesion, especially when used with AP134 industrial adhesion promoter. Both polyurethanes showed an order of magnitude greater adhesive strength compared to Sylgard-184 when cast in place.

Molding

Using techniques developed for casting PDMS microfluidic channels, microfluid channels were cast in PMC121 polyurethane elastomer using a positive photoresist mold. Similar to the Sylgard-184, the PMC121 was able to reproduce micro-scale channels. However, when inlet and outlet ports were punched the PMC121 did not exhibit the tendency to crack that is so problematic with PDMS. The flexibility and ability of the PMC121 to conform and seal to surfaces made it superior to the stiffer PMC790 for microfluidic channels. However, without AP134 or being cast in place, the PMC121 exhibited reduced adhesion to the substrate. Initial tests showed that the PMC121 was able to flow a dyed water solution at low pressure without leakage. The ability of PDMS channels to allow vapor to escape and thus enable deadhead channel filling has not been verified in polyurethane channels. Additional tests A number of additional tests were also carried out. Spin casting of the polyurethane elastomers on a traditional photoresist (PR) spinner was not successful. The polyurethanes would coalesce after spinning, creating voids in the resultant film. Oxygen plasma RIE etching was also found to not etch the polyurethanes appreciably. Metal deposition on polyurethanes was successfully performed using both Al and Au. These metals were then patterned via PR patterning and wet etching. Performing PR lithography for lift-off directly on the cured polyurethane was not successful. Upon development, the PR could not be fully patterned, indicating a possible chemical reaction with the polyurethane or an optical effect causing incomplete exposure of the bottom PR layer. These tests utilized AZ 5214 EIR, and other PR’s may be more compatible.

Summary and Conclusions

The work conducted in the past five years have met the project goals in all areas and exceeded the expectations in some. For example, a number of new materials for MEMS have been discovered and implemented. The top achievement in this effort has been:

- The successful development of three generations of artificial haircells and characterization of performances;

- The successful implementation of arrayed flow sensors on integrated circuit substrates with signal processing capabilities;

- The successful characterization of the artificial haircell sensors and match of performance to modeling;

- The direct comparison of biological haircells and artificial haircells in flow imaging of simple events (dipole movement). This makes way for future development of prototype sensors of larger scale. Certain industrial applications of the sensor arrays have become immediately possible, such as low cost bidirectional flow sensors for HVAC system monitoring. The results that have been obtained so far clearly point to the fact that haircells with sophisticated functions and much improved robustness can be made and successfully used by the NASA.

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